CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The balanced equation for burning of magnesium ribbon in air is: 1

(a) $Mg + O \longrightarrow MgO$ **(b)** $Mg + O_2 \longrightarrow MgO$ **(c)** $2\,Mg + O_2 \longrightarrow 2\,MgO_2$ **(d)** $2\,Mg + O_2 \longrightarrow 2\,MgO$

2. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 4 **(b)** 1 **(c)** 10 **(d)** 7

3. Which of the following pairs will give a displacement reaction? 1

(a) $FeSO_4$ solution and silver metal **(b)** $MgCl_2$ solution and aluminium metal

(c) NaCl solution and copper metal **(d)** $Fe + CuSO_4$ solution

4. What would be the electron dot structure of carbon dioxide (CO_2)? The number of shared electron pairs in it is: 1

(a) 4 **(b)** 2 **(c)** 6 **(d)** 3

5. Which of the following is a double displacement reaction? 1

(a) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ **(b)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ **(c)** $C + O_2 \longrightarrow CO_2$ **(d)** $CaCO_3 \longrightarrow CaO + CO_2$

6. The pH of gastric juice is about: 1

(a) 5.5 **(b)** 10 **(c)** 7.4 **(d)** 1.4

7. The reactivity of sodium is explained by: 1

(a) its hardness **(b)** its high melting point

(c) its easy loss of an electron **(d)** its large mass

8. Soaps are sodium or potassium salts of: 1

(a) aldehydes **(b)** short-chain alcohols

(c) alkanes **(d)** long-chain carboxylic acids

9. $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ is a displacement reaction because: 1

(a) zinc is more reactive and displaces copper from its salt

(b) zinc is less reactive than copper

(c) both metals are exchanged as anions

(d) copper gets oxidised

10. Milk of magnesia, an antacid, is: 1

(a) magnesium hydroxide **(b)** magnesium chloride

(c) magnesium carbonate **(d)** magnesium sulphate

11. Metals like sodium and potassium are extracted from their ores by: 1

(a) reduction with carbon **(b)** electrolysis of molten chlorides

(c) roasting **(d)** heating alone

12. The property of carbon to form long chains by bonding with itself is called: 1

(a) catenation **(b)** allotropy **(c)** polymerisation **(d)** isomerism

13. In the reaction $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$, the type of reaction is: 1

(a) Displacement **(b)** Double displacement

(c) Combination **(d)** Decomposition

14. A solution reacts with crushed egg-shells to give a gas that turns lime water milky. The solution contains: 1

(a) NaCl **(b)** HCl **(c)** NaOH **(d)** Na_2SO_4

15. Which of the following methods is suitable for preventing an iron frying pan from rusting? 1

(a) applying grease **(b)** applying paint
(c) applying a coating of zinc **(d)** all of these

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why should a magnesium ribbon be cleaned before it is burnt in air? 2

17. Why does dry HCl gas not change the colour of dry litmus paper? 2

18. Give an example of a metal which 2
(i) is a liquid at room temperature and
(ii) can be easily cut with a knife.

19. Why does carbon form compounds mainly by covalent bonding? 2

20. Why do we apply paint on iron articles? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why does the colour of copper sulphate solution change when an iron nail is dipped in it? Write the equation. 3

22. Write the general reaction of an acid with a metal. Give the reaction of dilute HCl with zinc and the test for the gas. 3

23. A hammer, a battery, a bulb, wires and a switch are given. How will you distinguish between metals and non-metals using them? 3

24. Draw the structures of the two isomers of butane and name them. 3

OR

What is a balanced chemical equation? Balance $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$. 3

25. What is the chlor-alkali process? Name its three products and write the equation. 3

OR

How is zinc obtained from its carbonate ore? Write equations. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. What is a balanced chemical equation? Balance $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$ step by step and write state symbols. 5

OR

Describe the reaction of dilute HCl with zinc granules. State the observations, the test for the gas, the equation and the general reaction. 5

27. Explain with equations the reactions of metals with oxygen, giving Na, Mg, Al and Cu as examples. State which oxide is amphoteric. 5

OR

Explain the bonding in carbon: four valence electrons, covalent bond, catenation and tetravalency. Draw dot structure of methane. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND